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MILITARY SPECIFICATION

CASTERS, INDUSTRIAL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers plate-mounted, individual-type truck casters.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard
PPP-B-576	Boxes, Wood Cleated, Veneer, Paper Overlaid
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Shipping, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed And Lock-Corner
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-B-676	Boxes, Setup

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ AFLC CASO/LODS, Federal Center, Battle Creek MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5340

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Military

MIL-P-116	Preservation-Packaging, Methods Of
MIL-F-3541	Fittings, Lubrication
MIL-L-10547	Liners, Case, And Sheet, Overwrap, Water-Vaporproof Or Waterproof, Flexible

STANDARDS

Federal

601	Rubber, Sampling And Testing

Military

MIL-STD-100	Engineering Drawing Practice.
MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-143	Standards And Specifications, Order of Precedence For The Selection Of
MIL-STD-808	Finishes, Protective, And Codes, For Finishing Schemes For Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation Of
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, And Waterproofing, With Appropriate Test Methods
MS24380	Caster, Industrial

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classifications

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago IL 60606.)

3. REQUIREMENTS

3.1 Qualification. The casters furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

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3.2 Components. The casters shall consist of the following major components:

Item Name	See Requirement
Mounting plate	3.8
Horn Assembly	3.9
Kingpin	3.10
Axle	3.11
Bearings	3.12
Wheels	3.13

3.3 Selection of specification and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials.

3.4.1 Protective treatment. Materials used in the construction of casters shall be protected against deterioration arising from environmental conditions of use, storage, and exposure. The protective media shall be stable and durable (shall not crack, chip, flake, powder, or scale) and shall not interfere with the performance characteristics required by this specification.

3.4.2 Metals. Metals shall be of the corrosion-resistant type or shall be protected to resist corrosion during normal service life.

* 3.4.2.1 Dissimilar metals. Insofar as practicable, the use of dissimilar metals in contact with each other shall be avoided. However, metal plating or metal spraying of dissimilar base metals to provide similar or suitable abutting surfaces will be permitted. The use of dissimilar metals separated by suitable insulating material will also be permitted. Dissimilar metals are defined in MIL-STD-889.

3.5 Design. The casters shall be designed for use as a running gear on equipment items that must be towed or moved on paved areas. The casters shall be stable when their housing is towed or pushed and so designed that vibration and shimmy are effectively damped out.

3.5.1 Maintainability. The casters shall be so designed and constructed that adjustments, repairs, and overhaul are held to a minimum, and when required, may be easily accomplished with a minimum of special tools.

3.6 Construction. The casters shall be so constructed that no parts will work loose in service. They shall be designed to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.6.1 Reliability. The casters shall be so designed and constructed as to insure the highest degree of operational reliability under the conditions specified herein.

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3.6.2 Sizes. Both rigid and swivel casters shall be constructed in accordance with MS24380 and shall be of the following sizes:

a. 4-inch

b. 6-inch

c. 8-inch

d. 10-inch

e. 12-inch.

3.6.3 Operational load capacities. The casters shall be constructed for sustained use under the rated loads specified on MS24380.

3.6.3.1 Shock-mounted wheels. Casters having shock-mounted wheels shall be capable of sustained use with the loads specified on MS24380, irrespective of spring rate variables and loading characterisitics peculiar to the shock-absorbing mechanism employed.

3.6.4 Static load. The casters shall be constructed to withstand a static load equal to four times the rated load specified on MS24380.

3.6.4.1 Shock-mounted wheels. Casters with shock-mounted wheels shall be constructed to absorb shock loads up to four times the rated load. Shock-mounted wheels, linkages, and mechanisms shall not hit mechanical stops or the maximum point of shock absorption when subjected to the loads specified herein.

3.6.5 Operation. Caster wheels shall rotate freely when operated at either no load or rated load. On swivel casters, the horn assembly shall swivel freely. There sh .ll be no indication of binding or unnecessary looseness in the casters.

3.6.5.1 When operating under full rated load or at no load, shock-mounted casters shall operate as specified in 3.6.3 when the shock absorbing mechanisms are disengaged.

3.6.6 Electrostatically conductive casters. Electrostatically conductive casters shall be designed and constructed to insure a positive electrical contact with the equipment they support. The wheel, tread bases, and cores shall be of such material and construction that the average electrical resistance of a number of measurements will be less than 250,000 ohms. Each measurement shall be taken from a different point at intervals on the periphery of the caster tread to the mounting plate. The maximum individual electrical resistance reading will be less than 1,000,000 ohms. Electrostatically conductive wheels shall be plainly and permanently marked on each side with the word "conductive" or other noticeable markings to permit positive identification so that electrostatically conductive wheels cannot be confused with standard wheels.

3.6.7 Floor markings. The wheel tread shall not leave floor markings, scuffing, or indentations of a permanent nature on a smooth hardwood or smooth, dry, troweled concrete surface.

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3.7 Performance.

3.7.1 Environmental. All casters shall be capable of operating in temperatures ranging from -20° to +120° Fahrenheit (F).

3.7.2 Endurance. When loaded to rated capacity, the casters shall satisfactorily withstand 60 hours of continuous operation over a smooth, uninterrupted running surface at 3 miles per hour (mph) for 4-inch casters, 5 mph for 6-inch casters, and 8 mph for 8-, 10-, and 12-inch casters.

3.7.3 Obstacles. When loaded to rated capacity, the casters shall satisfactorily withstand 2 hours of continuous operation over a running surface interspersed with obstacles at every 6 feet of travel: 1/8-inch obstacles for 4- and 6-inch casters traveling at speeds of 3 and 5 mph respectively, and 1/4-inch obstacles for 8-, 10-, and 12-inch casters traveling at a speed of 8 mph.

3.8 Mounting plate. The mounting plate shall be made of steel, cast iron, malleable iron, aluminum, or magnesium alloy.

3.9 Horn assembly. The horn assembly shall be made of steel plate, cast or forged steel, aluminum, or magnesium alloy. The design and quality of the horn assembly shall be adequate to insure rugged and rigid construction. The horn legs shall be either an integral part of the horn plate or welded thereto. The horn assembly of swivel casters shall be assembled to the mounting plate to permit independent rotation around the vertical axis.

3.10 Kingpin. The kingpin for swivel casters shall be constructed of steel of sufficient diameter and quality to assure resistance to distortion. The kingpin shall be so secured that in swiveling it will not rotate in frictional contact with component members of the swivel bearing assembly.

3.10.1 Nonadjustable kingpin. Four-, six-, and eight-inch swivel casters shall have a nonadjustable kingpin designed and constructed to efficiently maintain the proper relationship of swivel bearing components for the life of the casters under normal operating conditions.

3.10.2 Adjustable kingpin. Ten- and twelve-inch swivel casters shall have adjustable kingpins designed and constructed with threading (NF-2), a slotted nut, and a cotter pin to permit fine adjustment or replacement of the swivel unit bearing assemblies.

3.11 Axle. The axle shall be constructed of steel and shall secure the wheel by means of a nut. The axle design shall be such as to prevent the axle from revolving or the nut from becoming loose under normal working conditions.

3.11.1 The axle for casters employing commercial-type antifriction wheel bearings shall be designed to prevent clamping of the wheel between the legs of the horn assembly when the nut is drawn up tight on the axle bolt.

3.12 Bearings.

3.12.1 Swivel bearings. Swivel casters shall have a ball- or roller-bearing assembly so designed that in swiveling under load, all forces (component and direct thrusts or main and secondary loads) shall be efficiently transmitted through the antifriction bearings in contact with two completely effective sets of raceways.

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3.12.1.1 Ten- and twelve-inch swivel casters may have main load swivel bearings with raceways machined directed into the mounting and horn plate members, providing those members are constructed from at least a 5/8-inch thick steel plate and the ball bearings for this assembly are not less than 5/8 inch in diameter running in a raceway of not less than 7 inches in diameter to ball centers. The secondary swivel bearing shall be of a precision unit assembly having a life rating of 2,500 hours at 50 revolutions per minute (rpm) under offset and rated load and fabricated in accordance with high-grade bearing manufacturing practices. Optional designs of swivels for 10- and 12-inch casters may include both main and secondary swivel bearings as precision unit assemblies having a rating as shown above.

3.12.1.2 Four-, six-, and eight-inch casters shall have ball bearings or roller swivel bearings designed in accordance with the best standard of commercial antifriction swivel caster manufacturing practice.

3.12.1.2.1 Raceways, commercial-type ball or roller swivel bearings. The swivel bearing assembly raceways for 4-, 6-, and 8-inch casters shall have die-formed or machined raceways that are smooth and full-case hardened to a depth of not less than 0.015 inch after finish grinding.

3.12.1.3 Balls or rollers for swivel bearings. The balls or rollers of the swivel bearings for 4-, 6-, and 8-inch casters and main swivel bearings for 10- and 12-inch casters shall be constructed of steel, full-case hardened to the same minimum depths required for their respective raceways, and shall be ground and polished. The difference between any two diameters of the same ball or roller shall not exceed 0.005 inch. The difference between the diameters of any two balls or rollers in the same bearing assembly shall not exceed 0.001 inch. When precision unit assembly bearings are used for the primary thrust bearing of 10- and 12-inch casters, the bearing shall be in accordance with high-grade bearing manufacturing practice.

3.12.1.4 Hardness value of commercial-type swivel bearing surfaces. Swivel bearing surfaces shall be full-case hardened to the depths specified in 3.12.1.2.1 and 3.12.1.3, and to the hardness value specified herein to assure proper resistance to bearing surface wear, ball or roller failure, or ball or roller penetration of the raceway surfaces. All 4-, 6-, and 8-inch casters shall have swivel bearing surfaces of not less than 15N90 Rockwell hardness value, B scale.

3.12.2 Wheel bearings. Wheels for 4-, 6-, and 8-inch casters shall have commercial-type ball or roller bearings designed and manufactured in accordance with the best standard manufacturing practice for the intended purpose. Wheels for 10- and 12-inch casters shall have replaceable unit precision-type bearing assemblies having an average life rating of 2,500 hours at rated loads and rated speed and shall be manufactured in accordance with high-grade bearing manufacturing practice.

3.12.2.1 Raceways, commercial-type ball or roller wheel bearings. All raceways shall be hardened and shall assure contact with the ball, or line contact with the roller of bearings. For roller bearings, the inner raceway shall be part of or fitted on a full length seamless steel spanner bushing through which the axle is inserted. For ball bearings, the inner raceway shall butt in the wheel center to be a full length steel spanner bushing in effect, or have an equivalent means of eliminating the possibility of preloading and binding the bearings members when the axle is drawn up tight.

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3.12.2.2 Balls or rollers for wheel bearings. The ball or rollers of the wheel bearings shall be hardened and fabricated in accordance with the best bearing manufacturing practice for the purpose intended.

3.12.3 Lubrication. Unless permanent lifetime lubrication is furnished, all antifriction bearings shall be adequately lubricated upon assembly and shall be so mounted that they may be pressure lubricated through normal maintenance procedures. Wheel and swivel bearings shall have readily accessible, pressure-type lubrication fittings in accordance with MIL-F-3541.

3.12.4 Swivel lock. When specified by MS24380, a four-position, plunger-type-swivel lock shall be furnished on 4-, 6-, and 8-inch casters. When engaged, the swivel lock shall prevent the caster from swiveling while rolling or shall prevent rolling while at right angles to the normal path of the equipment. When disengaged, the swivel lock shall permit full 360 degrees free swiveling.

3.12.5 Miscellaneous bushing. Miscellaneous bushing for nonrotating (oscillating) bearing surfaces, peculiar to the linkages used in shock-mounted casters, shall be constructed of bearing bronze, steel-back bronze, or smooth, hardened steel.

3.13 Wheels. Caster wheels shall conform to the dimensions shown on MS24380 and to the requirements specified herein.

3.13.1 Shock mounting. When specified by MS24380, 8-, 10-, and 12-inch casters shall be provided with shock-mounted wheels capable of absorbing shock, vibration, and torsional stresses and shall preclude shimmy. Tracking shall permit vertical and longitudinal movement of the wheel within the caster horn assembly legs for the purpose of load, equipment, operating surface (floor), and running gear protection.

3.13.1.1 Deflection casters having shock-mounted wheels shall have equal heights under static rated loads within the limit specified on MS24380. Under static rated load, the amount of deflection shall be between a minimum of 15 percent and a maximum of 45 percent of the total amplitude of deflection which shall be 1 inch. The amount of deflection shall be measured using the center of the axle and the top plate as reference points.

3.13.2 Composition wheels (hard tread). Composition wheels shall be constructed with a tread of hard rubber or other composition and shall be of an all-molded construction or have a tread securely attached to the core or vulcanized to the core or tread base. The tread, tread base, and core shall have compressive, flexural, and impact strength sufficient to meet the requirements of 3.6 and 3.7. The treads shall be of nonmetallic construction and inherently sparkproof.

3.13.3 Rubber tread wheels. Rubber tread wheels shall have the cushion rubber tread securely vulcanized or cured-on to the nonstretchable core or tread base. The tread shall consist of manufactured or natural rubber, or any combination thereof, provided the material meets the requirements specified in 3.13.3.2.

3.13.3.1 Wheel cores and tread bases shall have compressive, flexural, and impact strength equal to the best standard practice for the intended purpose. Cores for wheels of 8-, 10-, and 12-inch casters shall be of one piece or bolted disc construction of fine grain cast iron, die formed steel, aluminum alloy, or magnesium alloy.

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3.13.3.2 Rubber tread physical properties.

3.13.3.2.1 Accelerated aging. The rubber shall be capable of resisting a temperature of 150°F for a period of 168 hours.

3.13.3.2.2 Hardness. The hardness of the tread shall be between 70 points minimum and 85 points maximum as measured on the face of the tread by a type A shore durometer.

3.13.3.2.3 Elongation. The elongation of the soft rubber treads shall be not less than 250 percent.

3.13.3:2.4 Tensile strength. The tensile strength of the tread shall be not less than 2,000 pounds per square inch (psi) on natural rubber and not less than 1,500 psi on manufactured rubber.

3.13.3.2.5 Compression set. The permanent set under full load deflection shall be not more than 50 percent of the deflection.

3.13.3.2.6 Adhesion. Soft treads shall not separate from their base or wheel rim when subjected to a force of 50 pounds per inch of tread width at the base of the tread.

3.13.4 Metal wheels. Metal wheels shall be of fine grain cast iron, steel, aluminum alloy, or magnesium alloy equal to the best standard practice for the intended purpose.

3.14 Brakes. When manually operated parking brakes are specified (see 6.2) for rigid or swivel casters, they shall be provided with control levers which are readily accessible for operation, permitting quick and easy brake application and release. Adjustable external brake shoes may be used providing they meet the performance specified herein.

3.14.1 Brake performance. Control levers shall be readily accessible for manual operation and shall not require a physical force in excess of 30 pounds to set the brakes. When the casters are loaded to the rated loads specified on MS24380, the brakes shall hold all caster wheels motionless whether headed up or down on a dry concrete 20 degrees incline.

3.15 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-STD-100 shall govern the changes in the manufacturer's part numbers.

3.16 Dimensions. The dimensions of the casters shall be as specified on MS24380.

3.17 Finishes and protective coatings. All exposed metal surfaces on caster assemblies that are subject to wear or corrosion shall be treated and painted in accordance with the requirements of MIL-STD-808.

3.18 Identification of product. Equipment, assemblies, and parts shall be marked foridentification in accordance with MIL-STD-130.

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3.19 Workmanship. The finished caster shall be clean, well made and free from defects which may affect appearance or serviceability.

3.19.1 Bolt connections. Bolt holes shall be accurately punched, drilled, reamed, or cast, and adjacent edges shall have no burrs or fins to interfere with proper seating of bolthead or nut. Where parts subject to critical vibrations are assembled with bolts or studs, nuts shall be sufficiently drawn against lock-washers, or castellated nuts shall be provided with cotter pins or safety wire of suitable size, or antivibration self-locking nuts that are properly drawn shall be used.

3.19.2 Welding. Welding shall be smooth, free from porosity or cracks or incomplete fusion, deformation of the metal, or rough and projecting edges. Where steel above 0.40 carbon content is welded in fabrication, a subsequent heat treatment for stress relief, or preheating, or low-hydrogen electrode shall be used to provide maximum physical properties in the joint.

3.19.3 Castings. Castings shall be well seasoned and shall be uniform in quality and condition free from injurious blowholes, porosity, hard spots, shrinkage, cracks, or other injurious defects. Castings shall not be repaired, plugged, or welded to affect adversely the strength, use, or machineability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of casters shall be as follows:

a.	Qualification testing	See 4.3
b.	Acceptance tests	See 4.4

4.3 Qualification testing (see 6.3).

4.3.1 Test samples. The test samples shall consist of two complete casters. Samples shall be appropriately identified with the manufacturer's part number and any additional identification required by the authorizing letter.

4.3.2 Test report. Upon completion of the qualification testing, a test report shall be prepared in accordance with MIL-STD-831 and three complete copies of the report furnished to the procuring activity.

4.3.3 Qualification tests. Qualification tests shall consist of all tests described under 4.5.

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4.4 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual test

b. Sampling tests.

4.4.1 Individual test. Each caster shall be subjected to an examination of product test as described under 4.5.

4.4.2 Sampling tests. A sample unit shall be one caster. A lot shall consist of all casters of the same Government designation offered for inspection by one manufacturer at one time. Sample units shall be selected in accordance with MIL-STD-105. Classification of defects shall be in accordance with Table I. The AQL to be employed shall be 4 percent for major defects and 10 percent for minor defects.

4.5 Test methods.

4.5.1 Examination of product. The casters shall be inspected to determine compliance with the requirements specified herein with respect to materials, finishes (including hardness), workmanship, and marking.

4.5.2 Dimensional check. The casters shall be dimensionally checked to determine compliance with the dimensions specified on MS24380.

4.5.3 Endurance test. The casters shall be loaded to rated capacity and shall be run at the following speeds on an endless belt or drum-type testing machine with a smooth, uninterrupted running surface for a period of 60 hours. At the completion of the test, there shall be no indications of excess wear to the bearing races, looseness of bearings, scuffing of tires, or looseness of tire tread. Any such indication shall be cause for rejection of the casters.

4-inch caster	3 mph
6-inch caster	5 mph
8-inch caster	8 mph
10-inch caster	8 mph
12-inch caster	8 mph.

4.5.4 Obstacle test. Each caster submitted for qualification shall be subjected to an obstacle test of 2 hours duration. The caster shall be loaded to its rated capacity and run at the speeds specified in 4.5.3 for the size indicated on an endless belt or drum-type testing machine for a period of 2 hours. There shall be an obstacle at every 6 feet of travel of the machine. At the completion of the test, there shall be no indication of excessive bearing wear, damage to bearing races, cutting or scuffing of tires, or looseness of tire tread. Any such indication shall be cause for rejection.

4.5.4.1 Obstacles for 8-, 10-, and 12-inch casters shall consist of a rectangular cross-sectioned steel bar 1/4 inch in thickness, so mounted that the longitudinal axis of the obstacles perpendicularly cross the path of the caster.

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TABLE I. Classification of defects.

Examine	Defect	Classification	
		Major	Minor
Finish	Not treated or painted as applicable		X
	Evidence of rust or corrosion	X	
Design	Dimensions not in accordance with MS24380	X	
	Any characteristic not in accordance with specified requirements	X	
Construction and workmanship	Not type and style of caster designated	X	
	Burrs or sharp edges	X	
	Improperly assembled	X	
	Bowed, bent, buckled, or otherwise deformed	X	
	Wheel bearings or swivel bearings bind when rotated	X	
	Tire loose on wheel	X	
	Tire contains blisters, pits, cracks porosity, gouges, checks, or discoloration	X	
	Threads stripped	X	
	Bolthead, washer, or nut does not seat properly due to fins or burrs or adjacent edges of bolt holes	X	
	Rivet heads not full, neatly made, concentric with rivet holes or in full contact with joining surface	X	
	Castings not free from blow holes, porosity, shrinkage, or cracks	X	
	Castings plugged or welded	X	
	Welds contain cracks, burnt holes, porosity, or incomplete fusion	X	
	Welds contain pits or fissures		X
Marking Identification	Missing, incomplete, not legible, not permanent, not located as specified		X

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4.5.4.2 Obstacles for 4- and 6-inch casters shall consist of a rectangular cross-section steel bar 1/8 inch in thickness, so mounted that the longitudinal axis of the obstacles perpendicularly cross the path of the caster.

4.5.4.3 The obstacles used on the endless belt or drum-type testing machine shall be replaced with new obstacles after each complete endurance test of casters having steel wheels.

4.5.5 Static test. A compressed load equal to four times the normal rated load shall be applied to the casters through the top plate by means of weights or a press for a period of 1 minute. The casters so tested shall be inspected for damage or permanent set as a result of the load applied. Damage or permanent set shall be cause for rejection.

4.5.5.1 Casters with shock-mounted wheels shall be tested under the same test loads and conditions specified in 4.5.5. The casters shall be tested with the shock-absorbing members engaged and operative and then with the members disengaged. The casters so tested shall be inspected to determine that no damage or permanent deformation has occurred. Damage shall be cause for rejection. This test shall be indicative of the caster strength and safety factor necessary to preclude the possibility of premature failure.

4.5.6 Floor marking. A test vehicle shall be equipped with four casters of the same wheel style and loaded with weights equal to four times the normal load rating for the individual caster less the weight of the dolly. The loaded test dolly shall be rolled back and forth on a smooth hardwood or smooth, dry, troweled concrete surface for 2 minutes. The surface shall be examined for any undue marks or scuffing. Markings, scuffing, or indentations of a permanent nature shall be cause for rejection.

4.5.7 Rubber tread test. The rubber tread on casters having this type tread shall meet the following tests:

4.5.7.1 Hardness. The hardness of the tread shall be between 70 points minimum and 85 points maximum as measured on the face of the tread by a type A Shore durometer.

4.5.7.2 Elongation. The elongation of the tread shall be not less than 250 percent when tested in accordance with Standard 601.

4.5.7.3 Tensile strength. The tensile strength of the tread shall be not less than 2,000 psi on natural rubber and not less than 1,500 psi on synthetic rubber when tested in accordance with Standard 601.

4.5.7.4 Accelerated aging. The rubber shall be subjected to a temperature of +158°F for a period of 166 hours by the air oven process specified in Standard 601. Upon completion of this test, the rubber shall be subjected to and satisfactorily pass the tests specified in 4.5.7.1, 4.5.7.2, and 4.5.7.3.

4.5.8 Compression set. The outside diameter of the wheel tread shall be measured within 0.010 inch and the point of measurement marked with a line not wider than 1/8 inch. The casters shall then be loaded to their normal rated load with the marked point of measurement at the point of ground contact and the tread deflection measured within 0.010 inch. The loaded casters shall then be placed in an air oven at 158°F for a period of 24 hours. At the end of this period, the load shall be removed and the casters allowed to cool at room

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temperature (approximately 70°F) for 30 minutes. The outside diameter of the wheel shall then be measured at the marked point. The percentage of set shall not exceed 50 percent when calculated as follows:

$$C = \frac{D - D_1}{R} \times 100$$

Where:

C = Percentage of set
D = Original diameter of wheel
D_1 = Final diameter of wheel
R = Tread deflection under load

4.5.9 Adhesion. Adhesion tests of the rubber tread molded to a metal or composition base shall be conducted at room temperature. The tread shall be cut transversely down to its base. The tread may be trimmed down to the base along the flanges to minimize the tearing of the tread. With the wheel horizontally supported on free rotating bearings in a suitable fixture, the tread shall be separated from the base circumferentially (by means of a testing machine exerting a radial pull) at the rate of approximately 6 inches per minute. The force necessary to separate the tread shall be not less than 50 pounds per inch of tread width at the base of the tread.

4.5.10 Electrostatic conductivity. The surface of the tread of electrostatically conductive casters shall be cleaned by any method which removes wax and dirt but does not abrade or permanently change the tread surface. After cleaning, the surface of the tread shall be dried and the caster shall be conditioned at room temperature with a relative humiidty of less than 80 percent for at least 24 hours. The caster shall be tested while supporting a load equal to 25 percent of its normal load rating. The caster shall be rolled onto a clean dry, flat metal plate. The metal plate and the caster mounting plate shall form the electrodes for the test. If the caster is tested while supporting equipment, all other casters under the vehicle shall be insulated from the floor on which the metal electrode plate is placed. The resistance between the electrodes shall be measured by any resistance measuring apparatus of suitable range which has an open circuit direct voltage (direct current) of approximately 500 volts (V). For the safety of the operator, the maximum current which can be delivered by the apparatus through a resistance of 500 ohms shall be less than 10 milliamperes (ma). Measurement shall be made with five separate areas of the tread successively in contact with the electrode plate and the average maximum values shall be determined. Average resistance for the five readings shall be less than 250,000 ohms and the maximum individual electrical resistance reading shall be less than 1,000,000 ohms.

4.5.11 Brake tests. When rigid or swivel casters are equipped with brakes, they shall be tested as follows:

a. Brake performance - The casters shall be loaded to rated loads as specified on MS24380 and the brakes applied. The physical force required on the control lever to set the brakes shall not exceed 30 pounds. The brakes shall hold the caster wheels motionless whether headed up or down on a dry concrete, 20 degree incline.

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b. Environmental effects - The casters with the brakes installed shall be subjected to each of the environmental tests specified in 4.5.12. At the conclusion of each test, the brakes and actuating system shall be examined. Any distortion, band or shoe separation, or other damage shall be cause for rejection. The casters shall then be subjected to the applicable performance test above. Any tendency toward erratic function, failure to pass, or other damage shall be cause for rejection.

4.5.12 Environmental tests. The following tests shall be conducted in accordance with the specified procedures of MIL-STD-810:

4.5.12.1 High temperature. The casters shall be subjected to the test specified in procedure II, except the temperature shall be +120°F. At the completion of the soaking period and with the temperature maintained, the casters shall be subjected to and satisfactorily pass the test specified in 4.5.4 herein.

4.5.12.2 Low temperature. The casters shall be subjected to the test specified in procedure I, except the temperature shall be -20°F. At the completion of the soaking period and with the temperature maintained, the casters shall be subjected to and satisfactorily pass the test specified in 4.5.4 herein

4.6 Inspection of the preservation, packaging, and packing. The inspection of the preservation, packaging, and packing shall be in accordance with the instructions in Section 5 herein.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Casters shall be preserved and packaged in accordance with method III of MIL-P-116. The unit containers shall conform to PPP-B-566, PPP-B-636, or PPP-B-676. Unit quantities shall be as specified by the procuring activity.

5.1.1.1 Intermediate packaging. Only identical casters shall be included in an intermediate package. Casters preserved and packaged in accordance with 5.1.1 shall be packaged in containers conforming to PPP-B-636. Unless otherwise specified by the procuring activity (see 6.2), the quantity of unit packages to be included in each intermediate package shall be at the option of the contractor and as governed by the limitations of the container being used.

5.1.2 Level C. Casters shall be packaged in a manner which provides adequate protection against physical damage during shipment from the supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice provided the latter meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2)

5.2.1 Level A. Casters shall be packed in shipping containers conforming to PPP-B-576, class 2; PPP-B-585, class 2 or 3; PPP-B-591, class 2; PPP-B-601, overseas type; PPP-B-621, class 2 PPP-B-636, class weather resistant; PPP-B-640, class 2. As far as practical, exterior containers shall be of uniform shape and size, of minimum cube and tare consistent with the protection required, and shall contain identical quantities. The gross weight of

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wood, wood-cleated, and triple wall boxes shall be limited to approximately 200 pounds. The gross weight of single wall and double wall fiberboard boxes shall be governed by the weight limitation of the box specifications. Containers shall be closed and strapped in accordance with the applicable container specification and the appendix thereto. Wood and wood-cleated boxes shall be provided with a case liner conforming to MIL-C-10547 and shall be sealed in accordance with appendix thereto. Fiberboard shipping containers shall be waterproofed or sealed in accordance with the appendix of the applicable container specification.

5.2.2 Level B. Level B pack shall be the same as level A, except the container shall conform to PPP-B-601, domestic type, style A or B.

5.2.3 Level C. Packages which require overpacking for acceptance by the carrier shall be packed in exterior shipping containers in a manner that will insure safe transportation, at the lowest rate, to the point of delivery. Containers and packing shall comply with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.3 Physical protection. Cushioning, blocking, bracing, and bolting as required shall be in accordance with MIL-STD-1186, except for domestic shipments, waterproofing requirements for cushioning materials and containers shall be waived. The drop tests of MIL-STD-1186 shall be waived when preservation, packaging, and packing of the casters are for immediate use or when the drop tests of MIL-P-116 are applicable.

5.4 Marking. In addition to any special marking required by the contract or order, the unit, intermediate, and shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

CASTER, INDUSTRIAL, RIGID OR SWIVEL (as applicable).

6. NOTES

6.1 Intended use. The casters covered by this specification are intended for use in accordance with their respective duty and operational load ratings.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. When manually operated parking brakes are required (see 3.14).

c. MS part number (see MS24380).

d. Selection of applicable levels of preservation and packaging, and packing (see 5.1 and 5.2).

e. Unit quantities for level A packaging (see 5.1.1).

f. When applicable, the quantity of unit packages to be included in intermediate packages for level A packaging (see 5.1.1.1).

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* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is HQ AFLC CASO/LODS, Federal Center, Battle Creek MI 49016, and information pertaining to qualification of products may be obtained from that activity.

6.4 Definitions.

6.4.1. Bearings.

6.4.1.1 Commercial antifriction swivel. Bearings of this type consist of those assemblies considered as nonreplaceable, using balls or rollers of uniform size operating in contact with raceways formed directly into major parts of the caster assembly.

6.4.1.2 Commercial antifriction wheel. Bearings of this type consist of balls or rollers of uniform size operating in contact with raceways. They will be made in accordance with the best standard of commercial practice for the purpose intended.

6.4.1.3 Precision type. Bearings of this type consist of unit assemblies made to close tolerances in accordance with high-grade bearing manufacturing practice which will be replaceable as complete units.

6.4.1.4 Surface hardness, value of. Because of the desirability of having tough, shock-resistant inner physical characteristics of antifriction bearing raceways, the outer surfaces will be fullcase hardened. High-surface hardness values and good depth of hardness offer resistance to ball or roller penetration and wear of antifriction bearing raceway surfaces.

6.4.2 Bushings.

6.4.2.1 Miscellaneous. These bushings are of the nonrotating, oscillating type as employed in linkages of shock-mounted casters and are intended to reduce wear at these points of friction.

6.4.2.2 Spanner. A spanner bushing is a seamless steel sleeve through which the axle bolt passes and around which the wheel revolves. The spanner bushing (or its effective equivalent) acts as an inner bearing surface when used with antifriction bearings. The bushing maintains proper horn assembly leg spacing thus strengthening thy horn assembly by tying the horn legs together when the axle nut is drawn up tight.

6.4.3 Caster. The caster is the finished product as covered by this specification.

6.4.3.1 Caster assembly. The caster assembly is comprised of all component parts of a caster, less only the wheel and spanner bushing. The axle bolt and nut are to be supplied as a part of, but not assembled to the caster assembly.

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6.4.3.2 Rigid. A rigid caster is a nonsviveling, fixed caster permitting horizontal rolling of the wheel in only forward or reverse direction.

6.4.3.3 Swivel. A swivel caster is a caster designed with swivel bearings and swivel offset permitting full 360 degree horizontal rotation of the horn assembly and wheel around the vertical axis (kingpin) of the swivel bearing assembly while the adapter (flat top mounting plate, et cetera) is held in a fixed position. These free swiveling casters permit easy maneuverability of the vehicle (equipment) through applied force.

6.4.4 Electrostatic conductivity. Electrostatic conductivity is the ability of a caster to transmit and dissipate electrostatic charges that build up in a vehicle.

6.4.5 Overall height. The overall height is the vertical distance between the top of the caster mounting plate and the bottom of the wheel which in effect is the vertical distance between the floor surface and the equipment base.

6.4.6 Horn assembly. The horn assembly is the unit comprising the horn plate and the legs.

6.4.6.1 Horn plate. The horn plate is the horizontal member at the top of the horn legs.

6.4.6.2 Kingpin. The kingpin comprises the means of efficiently maintaining the proper relationship of the swivel bearing parts for the life of the caster under normal operating conditions.

a. Adjustable kingpin - An adjustable kingpin is for use with swivel bearing assemblies which, with the kingpin, are designed to permit compensat ing adjustment for wear, et cetera.

b. Nonadjustable kingpin - A nonadjustable kingpin is for use with commercial antifriction swivel bearing assemblies where the bearing assemblies are not replaceable units and are designed to perform efficiently without adjustment for the normal life of the caster.

6.4.6.3 Legs. The legs are that part or parts of a caster used as a means of supporting the wheel of a conventional caster.

6.4.7 Linkages. Linkages are those parts of a shock-absorbing caster to which the wheel load axle and shock-absorbing mechanisms are attached thus permitting the wheel to properly function and deflect when hitting obstacles.

6.4.8 Load.

6.4.8.1 Shock. Shock load is the momentary exertion of maximum force against the caster resulting from extreme vibration or impact shock as encountered in operation.

6.4.8.2 Static. Static load is the load imposed on a caster or wheel not in motion.

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6.4.9 Mounting plate. The mounting plate is the part of a caster used as a means to attach or adapt the caster to equipment.

6.4.10 Load rating. The load rating is a generally recommended operational load rating for normal operating conditions, expressed in pounds per caster. Load ratings will reflect an allowance for a reasonable safety overload factor.

6.4.11 Caster size. The caster size will be the nominal diameter of the wheel.

6.4.12 Exposed surface. The exposed surface is the metal surface which, after assembly, is exposed to corrosion or wear. Swivel and wheel bearing surfaces, miscellaneous bushing surfaces, and self-lubricating bearing surfaces are not considered exposed surfaces.

6.4.13 Tread width. Tread width is the width of the wheel tread (tire) as measured at the base of the tread proper.

6.4.14 Shock-mounted wheel. A shock-mounted wheel is a wheel provided with a mechanical means to absorb impact shock forces, vibration, and stresses induced by operating surface (floor) conditions. This shock mount construction protects loads, vehicles, and operating surfaces as well as improving the operating characteristics and service life of the caster or wheel.

6.4.15 Sparkproof wheel (nonsparking). Sparkproof wheels are those which contain a composition or rubber tread material that will withstand friction without sparking.

6.5 Caster selection. In selecting a caster, consideration should be given to the following:

6.5.1 The type of duty and loads to be carried should be the determining factor in selecting the caster. Note that load ratings shown on MS24380 are recommended operational load ratings in keeping with the strength and performance characteristics of the caster assembly and wheel. These recommended load ratings should be adhered to in order to obtain the best service life of the caster.

6.5.2 The floor or operating surface conditions, weight and kind of load to be carried (see 6.1), desirable wheel tread materials, and caster duty classifications should be carefully considered.

6.5.3 Rough operating surfaces or those having other shock inducing obstacles such as door sills dictate the need of wheels capable of absorbing these shocks while permitting easy rolling and maneuverability. Rubber tread wheels afford some cushioning of shocks and are generally more floor protecting than other tread materials. Shock-mounted wheels are capable of absorbing damanging shock, vibrations, and torsional stresses and should be considered where the protection of easily damaged loads, equipment, and operating surfaces are of importance.

6.5.3.1 Where quietness of operation is to be considered, the rubber tread wheels are recommended. Heavy loads are best supported on metal tread wheels; however, these wheels are not as floor protective as rubber tread wheels.

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6.5.3.2 High heat and normal floor concentrations of acids or alkalies are coped with by composition wheels which are capable of resisting the destructive action of water, oil, normal floor concentrations of mild acids and floor cleaning compounds.

6.5.3.3 Metal wheels are not considered as floor protective and should be used only in cases where absolutely necessary. In lieu of metal wheels, due consideration should be given to composition wheels where, due to higher operational load ratings, the rubber tread wheels cannot be used.

6.5.4 Where explosion hazards exist, the composition or rubber tread wheels should be considered due to their inherent sparkproof nature. Electrostatically conductive casters should also be considered; however, the use of electostatically conductive casters is cautioned in that the floor in such areas must also be conductive, and that other apparatus must rest on conductive leg tips, glides, or equivalent. Note also that waxed floors will nullify the conductivity characteristics of casters.

* 6.6 Reclaimed materiels. The use of reclaimed materiels shall be encouraged to the maximum extent possible.

6.7 Changes from previous issue. The margins of this specification are marked with an asterisk to indcate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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